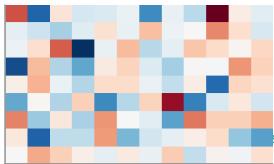


A Genome-scale screen → measured effect vectors



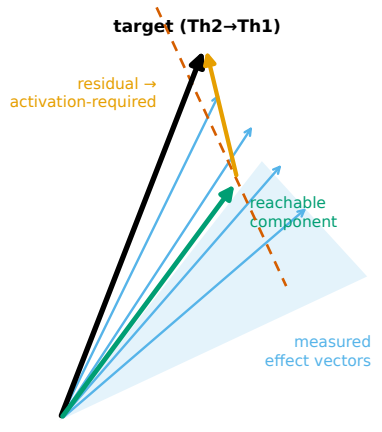
CRISPRi Perturb-seq
primary human CD4⁺ T cells



33,983 knockdowns × 10,282 genes

each column =
a measured
effect vector

B Reachability = membership in a convex cone



non-negative combination (NNLS); “outside”
certified by a separating hyperplane (Farkas / KKT)

C Verdict + decomposition + activation certificate (Th2→Th1)

signed decomposition of the target



LOF (reachable)

GOF (activation)

neither

verdict: partly reachable

held-out cosine 0.448 · null z ≈ 24 · ||KKT/Farkas|| 1.1 × 10⁻¹¹

activation certificate — genes that must be raised

(no knockdown mix delivers them):

● LYAR ● IKZF3 ● CRTAM ● CBLB
● GBP5 ● LAG3 ● IRF8

positive control GATA3↓ reached (rank 155/6871)

D Same operator transfers unchanged

Norman 2019 · K562

CRISPR activation (gain-of-function)
held-out target: CEBPA state



held-out cosine 0.878

null z ≈ 37 · held-out-gene z 23.5
recipe lead: CEBPE (CEBPA paralog)

additivity median cos 0.71

(bounded approximation, tested)

**different screen, direction, cell type —
same convex-cone operator, no retuning**